# Time Series Analysis

## Chapter 01
## Introduction

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- Sum of Random Variables

# What are Time Series Data?

- Time series data are collected sequentially over time.
- Some common example include:
  1. Meteorological data (temperatures, precipitation levels, etc.) taken daily or hourly.
  2. Sales data taken annually
  3. Heart activity measured at each millisecond
- The major goals of time series analysis are: (1) to model the stochastic phenomenon that produces these data; and (2) to predict or forecast the future values of the time series.
- A key aspect of time series data is that the observations measured across time are typically dependent random variables, not independent r.v.'s.

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# Example: Annual Rainfall in Los Angeles

- Annual rainfall amounts recorded in Los Angeles, California, over more than 100 years.

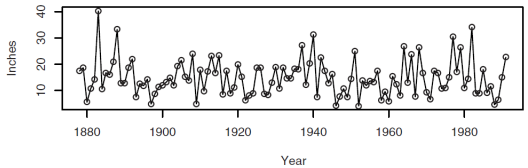



Figure: Time Series Plot of Los Angeles Annual Rainfall

- There is substantial variation in rainfall amount across years.
- Are consecutive year related? Can we predict a year's rainfall amount based on the previous year's amount?

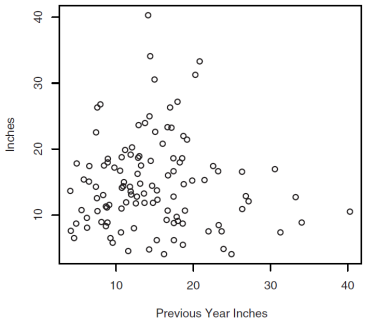



Figure: Scatterplot If LA Rainfall vs. Last Year's LA Rainfall

- Scatter plot of one year's rainfall against previous year's rainfall shows little association.

# R-code

```
library(TSA)

# Time Series Plot of Los Angeles Annual Rainfall
win.graph(width=4.875, height=2.5, pointsize=8)
data(larain); plot(larain, ylab='Inches', xlab='Year', type='o')

# Scatterplot of LA Rainfall vs. Last Year's LA Rainfall
win.graph(width=3, height=3, pointsize=8)
plot(y=larain, x=zlag(larain), ylab='Inches',
     xlab='Previous Year Inches')
```

# Example: An Industrial Chemical Process

- A time series from an industrial chemical process. The variable measured is a color property from consecutive batches in the process.

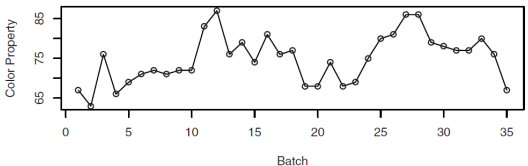



Figure: Time Series Plot of Color Property from a Chemical Process

- Can we predict a batch's color property based on the previous batch's value?

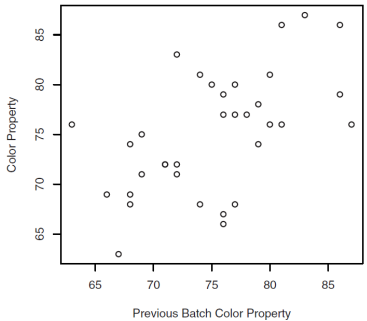



Figure: Scatterplot of Color Value vs. Previous Color Value

- Scatter plot of one batch's color value against previous batch's value shows some positive association.

# R-code

```
# Time Series Plot of Color Property from a Chemical Process
win.graph(width=4.875, height=2.5,pointsize=8)
data(color)
plot(color,ylab='Color Property',xlab='Batch',type='o')

# Scatterplot of Color Value vs. Previous Color Value
win.graph(width=3,height=3,pointsize=8)
plot(y=color,x=zlag(color),ylab='Color Property',
     xlab='Previous Batch Color Property')
```

# Example: Annual Abundance of Canadian Hare

- The annual abundance of Canadian hare.

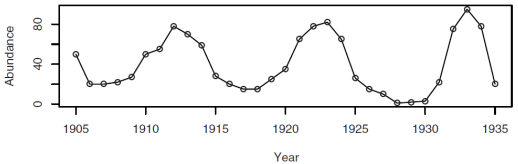



Figure: Abundance of Canadian Hare

- Can we predict a year's abundance based on the previous year's abundance?

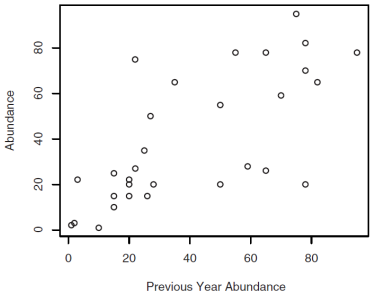



Figure: Hare Abundance vs. Previous Year's Hare Abundance

- Scatter plot of one year's abundance against previous year's abundance shows clear positive association.

# R-code

```
# Time Series Plot of Abundance of Canadian Hare
win.graph(width=4.875, height=2.5,pointsize=8)
data(hare); plot(hare,ylab='Abundance',xlab='Year',type='o')

# Scatterplot of Hare Abundance versus Previous Year's Hare Abundance
win.graph(width=3, height=3,pointsize=8)
plot(y=hare,x=zlag(hare),ylab='Abundance',
     xlab='Previous Year Abundance')
```

# Example: Average Monthly Temperature, Dubuque, Iowa

- The average monthly temperature (in degree Fahrenheit) in Dubuque, Iowa.

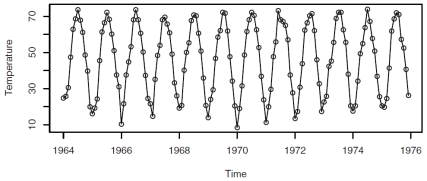



Figure: Average Monthly Temperatures, Dubuque, Iowa

- This plot notice the regular pattern.
- These data show seasonality: observations that are 12 months apart are related.
- Each January temperature is low while each July temperature is high.

# Monthly Oil Fitler Sales

- Monthly sales to dealers of a specialty oil filter.

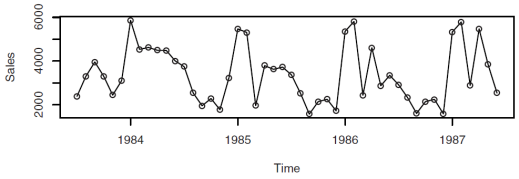



Figure: Monthly Oil Filter Sales

- A seasonal pattern is seen in the data.

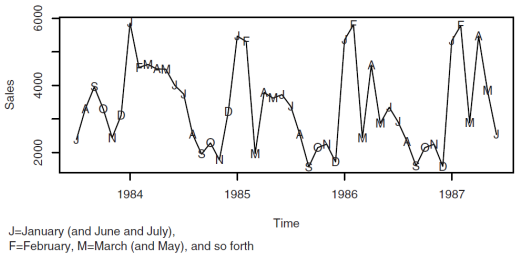



Figure: Monthly Oil Filter Sale with Special Plotting Symbols

- Certain months regularly see high sales while other months regularly see low sales.

# R-code

```
# Time Series Plot of Average Monthly Temperature
win.graph(width=4.875, height=2.5,pointsize=8)
data(tempdub); plot(tempdub,ylab='Temperature',type='o')

# Time Series Plot of Monthly Oil Filter Sales
data(oilfilters); plot(oilfilters,type='o',ylab='Sales')

# Time Series Plot of Monthly Oil Filter Sales
# with Special Plotting Symbols
plot(oilfilters,type='l',ylab='Sales')
points(y=oilfilters,x=time(oilfilters),
       pch=as.vector(season(oilfilters)))
```

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# A Model-Building Strategy

- Finding appropriate models for time series is a nontrivial task.
- There are three main steps to build a model:
  1. model specification (or identification)
  2. model fitting, and
  3. model diagnostics
- In model specification, we choose models based on the time series plots.
- In choosing a model, we shall follow the principle of parsimony.
- In model fitting, we shall consider the criteria such as least squares and maximum likelihood for estimates.
- In model diagnostics, we assess the quality of the model that we have specified and estimated.

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# Notation

- A set is a well-defined collection of objects called elements or members of the set. We use capital letters such as $A, E, S, \Omega, ...$ to denote a set and the lowercase such as $a, e, s, \omega, ...$ to denote the elements of the set.
  - E.g., the set of natural number is denoted by $\mathbb{N} = \{1, 2, 3, ...\}$.
- Empty set is a set that contains no element and is denoted by $\emptyset$. Hence, $= \{\}$.
- The set $E$ is called a subset of the set , written as $E \subset \Omega$, if each element of $E$ is also an element of $\Omega$. That is, $E \subset \Omega$ if $x \in E$, then $x \in \Omega$. E.g., $\mathbb{E} =$ set of even numbers. Then $\mathbb{E} \subset \mathbb{N}$.
- The poset of $\Omega$, denoted by $\mathcal{P}(\Omega)$, is the collection of all subsets of $\Omega$.
- A universal set is a set which contains all the elements or objects of other sets, including its own elements.

# Notation

- The complement of a set $E$ with respect to the universal set $\Omega$ is denoted by $E^c$ or $\overline{E}$. It includes all the elements of $\Omega$ that are not present in $E$.
  - E.g., If $\Omega = \{1, 2, 3, 4\}$ and $E = \{1, 3\}$, then $E^c = \{2, 4\}$.
- The union ($\cup$) of two or more sets is the set containing all the elements of each set. That is, $A \cup B = \{x : x \in A \text{ or } x \in B\}$.
  - E.g., If $E = \{1, 2, 3\}$ and $F = \{4, 5\}$, then $E \cup F = \{1, 2, 3, 4, 5\}$.
- The intersection ($\cap$) of two or more sets is the set containing only the common elements of each set. That is, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.
  - E.g., If $E = \{1, 2, 3, 4\}$ and $F = \{4, 5\}$, then $E \cap F = \{4\}$.

# Random experiment, sample space, $\sigma$-algebra

- A random experiment is an experiment whose outcome cannot be predicted with certainty.
  - For example, the value of a stock price can go up or down, and no one can be certain about its state. Here, the outcome can be "up" or "down".
- The sample space of a random experiment is the collection of all possible outcomes. It is often denoted by $S$ or $\Omega$. E.g., $\Omega = \{\text{up}, \text{down}\}$.
- An $\sigma$-algebra (also: $\sigma$-field) on the sample space $\Omega$ is a collection $\Sigma$ of subsets of $\Omega$ that:
  1. includes the empty set,
  2. is closed under complement, and
  3. is closed under countable unions and countable intersections.

The pair $(\Omega, \Sigma)$ is called a measurable space or Borel space.

# Random experiment, sample space, $\sigma$-algebra

- Mathematically, the collection $\Sigma$ of the subsets of the sample space $\Omega$ is a $\sigma$-algebra if:
  1. $\emptyset \in \Sigma$
  2. if $E \in \Sigma$, then $E^c \in \Sigma$
  3. if $E_1, E_2, ... \in \Sigma$, then $\bigcup_{n=1}^{\infty} E_n \in \Sigma$

Consider the $\sigma$-algebra $\Sigma$ as the collection of available information associated to the sample space $\Omega$ of the underlying random experiment.

- Each element of the $\sigma$-algebra $\Sigma$ is called an event.
  - Null event is $\emptyset$ and certain event is $\Sigma$.

# Random experiment, sample space, $\sigma$-algebra

- **Example**: Consider the sample space $\Omega = \{\text{up}, \text{down}\}$.
  Then $\Sigma = \{\emptyset, \{\text{up}\}, \{\text{down}\}, \Omega\}$ defines a $\sigma$-algebra on $\Omega$.

**Check**:

1. $\emptyset \in \Sigma$
2. If $E \in \Sigma$, then $E^c \in \Sigma$
3. If $E_1, E_2, \ldots \in \Sigma$, then $\bigcup_{n=1}^{\infty} E_n \in \Sigma$

# Random experiment, sample space, $\sigma$-algebra

- $\sigma$-algebra generated by a collection $\mathcal{F} \subset \mathcal{P}(\Omega)$, denoted by $\sigma(\mathcal{F})$, is the smallest $\sigma$-algebra that contain $\mathcal{F}$.
- **Example**: Consider the sample space $\Omega = \{\text{up}, \text{down}\}$. Then $\Sigma = \{\emptyset, \{\text{up}\}, \{\text{down}\}, \Omega\}$ define a $\sigma$-algebra. Let $\mathcal{F} = \{\{\text{up}\}\}$. Then $\Sigma = \sigma(\mathcal{F})$.

# Probability

- The function $\mathbb{P} : \Sigma \to [0, 1]$ is called probability function if:
  1. $\mathbb{P}(E) \geq 0$ for any event $E$
  2. $\mathbb{P}(\Omega) = 1$
  3. If $E_1, E_2, ... \in \Sigma$, such that $E_i \cap E_j = \emptyset$ for any $i \neq j$, we have $\mathbb{P}\left(\cup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(E_n)$
- The triplet $(\Omega, \Sigma, \mathbb{P})$ is called probability space.

# Probability

- **Inclusion**: If $E \subset F$, then $\mathbb{P}(E) \leq \mathbb{P}(F)$.
- **Addition Rule**: For any two events $E$ and $F$, we have

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F)$$

- **Probability of complete event**: For any event $E$, we have

$$\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$$

# Conditional Probability

- **Conditional probability** refers to the chances that some event ($E$) occurs given that another event ($F$) has also occurred, and is denoted by $\mathbb{P}(E|F)$. Mathematically, we define $\mathbb{P}(E|F)$ by:

$$\mathbb{P}(E|F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)},$$

  provided that $\mathbb{P}(F) \neq 0$.

- **Multiplicative Rule**: For two events $E$ and $F$, we have

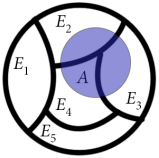
$$\mathbb{P}(E \cap F) = \mathbb{P}(E|F)\mathbb{P}(F)$$

# Law of Total Probability

- For **any partition** $E_1, E_2, ..., E_n$ of $\Omega$, that is $E_i \cap E_j = \emptyset$ for all $i \neq j$ and $\cup_{i=1}^{n} E_i = \Omega$ (in other word, we call $E_1, E_2, ..., E_n$ mutually exhaustive), we have

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap E_1) + \mathbb{P}(A \cap E_2) + ... + \mathbb{P}(A \cap E_n) \\ &= \mathbb{P}(A|E_1)\mathbb{P}(E_1) + \mathbb{P}(A|E_2)\mathbb{P}(E_2) + ... + \mathbb{P}(A|E_n)\mathbb{P}(E_n)\end{aligned}$$

# Law of Total Probability




$$
\begin{aligned}
P(A) &= P(E_1 \cap A) + P(E_2 \cap A) + P(E_3 \cap A) + P(E_4 \cap A) + P(E_5 \cap A) \\
&= \sum_{i=1}^{5} P(E_i \cap A) \\
&= P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + P(A|E_3)P(E_3) + P(A|E_4)P(E_4) + P(A|E_5)P(E_5) \\
&= \sum_{i=1}^{5} P(A|E_i)P(E_i)
\end{aligned}
$$

# Law of Total Probability: Example

- A quick example of the application of the rule of total probability.
- Whether or not I go hiking depends on the weather, if it is sunny there is a 60% I will go for a hike, while there is only a 10% chance if it is raining and a 30% chance if it is snowing.
- The weather forecast for tomorrow calls for 50% chance of sunshine, 40% chance of rain, and a 10% chance of snow.
- What is the probability I go for a hike tomorrow?

# Independence

- We say that two events $E$ and $F$ are **independent** if

$$P(E \cap F) = P(E)P(F)$$

which also implies that

$$P(E|F) = P(E)$$
$$P(F|E) = P(F)$$

- This should not to be confused with disjoint (mutually exclusive) events where $P(E \cap F) = P(\emptyset) = 0$.

# Independence

- We say that events $E_1, E_2, ..., E_n$ are **independent** if for **every** subset $E_{i_1}, E_{i_2}, ..., E_{i_k}$,

$$P\left(\bigcap_{j=1}^{k} E_{i_j}\right) = \prod_{j=1}^{k} P(E_{i_j})$$

- Note that $P\left(\bigcap_{j=1}^{n} E_j\right) = \prod_{j=1}^{n} P(E_j)$ is **not** sufficient for independence.

# Independence: Example

**Table 3.3.1** Hair color and eye color

| | | Hair color | | | |
|---|---|---|---|---|---|
| | | Brown | Black | Red | Total |
| Eye color | Brown | 400 | 300 | 20 | 720 |
| | Blue | 800 | 200 | 50 | 1,050 |
| | Total | 1,200 | 500 | 70 | 1,770 |

1. Are brown and black hair disjoint?
2. Are brown and black hair independent?
3. Are brown eyes and red hair disjoint?
4. Are brown eyes and red hair independent?

# Bayes' Formula

- Theorem: For two events $E$ and $F$ with $P(E), P(F) \neq 0$, we have

$$P(F|E) = \frac{P(E|F)P(F)}{P(E)}.$$

Using Total Probability Law, we have

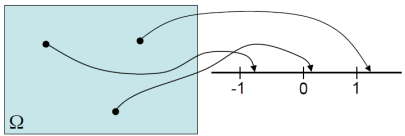
$$P(F|E) = \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|F^c)P(F^c)}.$$

# Bayes' Theorem: Example

- If you've ever watched the TV show *House*, you know that Dr. House regularly stats, "It's never lupus".
- Lupus is a medical phenomenon where antibodies that are supposed to attack foreign cells to prevent infections instead see plasma proteins as foreign bodies, leading to a high risk of blood clotting. It is believed that 2% of the population suffer from this disease.
- The test for lupus is very accurate if the person actually has lupus, however is very inaccurate if the person does not. More specifically, the test is 90% accurate if a person actually has the disease. The test is 74% accurate if a person does not have the disease.
- Is Dr. House correct event if someone test positive for Lupus?

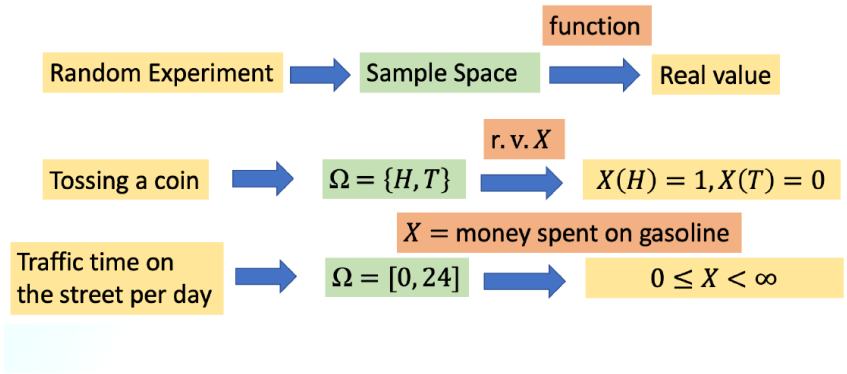
# Definition of Random Variable

- A **random variable** is a function or rule that assigns a numerical value to each outcome in the sample space of a random experiment.
- We use $X$, hence $X : \Omega \to \mathbb{R}$, when referring to a random variable in general, while specific values of $X$ are shown in lowercase (e.g., $x$). The image of $X$ is denoted by $X(\Omega)$.

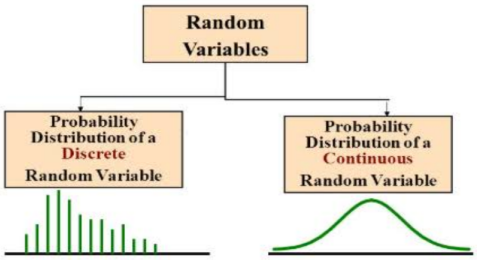



- The random variable often is a direct result of an observational experiment (e.g., counting the number of takeoffs in a given hour at Phnom Penh International Airport).

# Random Variable: Example



# Type of Random Variables




- A **discrete random variable** has a countable number of distinct values.
- A **continuous random variable** assume any value on some interval of the set of real numbers.

# Type of Random Variable

| *Decision Problem* | *Discrete Random Variable* |
|---|---|
| Oxford University has space in its MBA program for 65 new students. In the past, 75 percent of those who are admitted actually enroll. The decision is made to admit 80 students. What is the probability that more than 65 admitted students will actually enroll? | $X$ = number of admitted MBA students who actually enroll ($X = 0, 1, 2, ..., 80$) |
| On the late morning (9 to 12) work shift, L.L. Bean's order processing center staff can handle up to 5 orders per minute. The mean arrival rate is 3.5 orders per minute. What is the probability that more than 5 orders will arrive in a given minute? | $X$ = number of phone calls that arrive in a given minute at the L.L. Bean order processing center ($X = 0, 1, 2, ...$) |
| Rolled steel from a certain supplier averages 0.01 defect per linear meter. Toyota will reject a shipment of 500 linear meters if inspection reveals more than 10 defects. What is the probability that the order will be rejected? | $X$ = number of defects in 500 meters of rolled steel ($X = 0, 1, 2, ...$) |

# Probability Mass Function (PMF)

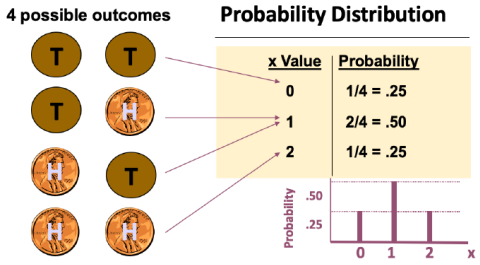
- When $X(\Omega)$ is countable, we call the r.v. $X$ to be **discrete**.
- We define the **probability mass function (PMF)** $P_X : X(\Omega) \to \mathbb{R}$ by:

$$P_X(x) = P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\})$$

- The PMF satisfies the following properties:
  - For all $x \in X(\Omega), 0 \leq P_X(x) \leq 1$
  - $\sum_{x \in X(\Omega)} P_X(x) = 1$
  - For any event $E \in \sum$, we have $P(X \in E) = \sum_{x \in E} P_X(x)$.

# Probability Mass Function: Example

- Two identical fair coins are tossed. Let $X$ = number of resulting heads (H).
- There are 4 possible outcomes, hence $\Omega = \{HH, HT, TH, TT\}$ and $X(\Omega) = 0, 1, 2$.




- Remark. "Probability distribution" refers the probability table, the graph or PMF.

# Cumulative Distribution Function (CDF)

- When $X(\Omega)$ is countable, we call the r.v. $X$ to be **discrete**.
- We define the cumulative distribution function (CDF) $F_X : \mathbb{R} \to \mathbb{R}$ by:

$$F_X(x) = P(X \leq x) = \sum_{k \leq x} P_X(x).$$

- The CDF satisfies the following properties:
  - For all $x \in \mathbb{R}, 0 \leq F_X(x) \leq 1$
  - $\lim_{x \to -\infty} F_X(x) = 0$
  - $\lim_{x \to +\infty} F_X(x) = 1$
  - $F_x(x)$ is increasing and right continuous.

# Cumulative Distribution Function (CDF)

- Example: Two identical fair coins are tossed. Let $X$ = number of resulting heads. Then $X(\Omega) = \{0, 1, 2\}$ and

$$P_X(0) = 1/4, P_X(1) = 1/2 \text{ and } P_X(2) = 1/4$$

  - If $x < 0$, then $F_X(x) = 0$
  - If $0 \leq x < 1$, then $F_X(x) = P_X(0) = 1/4$
  - If $1 \leq x < 2$, then $F_X(x) = P_X(0) + P_X(1) = 3/4$
  - If $x \geq 2$, then $F_X(x) = P_X(0) + P_x(1) + P_X(2) = 1$

# Mean or Expected Value

- The **mean** or **expected value** of the r.v. $X$, denoted by $E(X)$ or $\mu_X$, is given by

$$E(X) = \sum)x \in X(\Omega)xP_X(x).$$

- Example: Two identical fair coins are tossed. Let $X$ = number of resulting heads. Then

$$E(X) = \sum_{x \in X(\Omega)} xP_X(x) = 0 \times 1/4 + 1 \times 1/2 + 2 \times 1/4 = 1.$$

- Interpretation: On average, we get 1 head among two tosses.

# Mode

- The **mode** value of the r.v. $X$, denoted by mode $(X)$, is the value $x^*$, if any, such that $P_X(x)$ is maximized ($x^*$ appears the most frequently in the data). That is,

$$x^* = arg \max_{x \in X(\Omega)} P_X(x)$$

- Example: Two identical fair coins are tossed. Let $X =$ number of resulting heads. Then $x = 1$ has the highest frequency (that is $P_X(1)$ is the maximum). Hence mode $(X)$=1.
- Interpretation: The data value that occurs the most is $x = 1$.

# Median

- The **median** value of the r.v. $X$ is the value $\tilde{x}$ that divides the data into two equal parts (hence 50%-50%). That is, if $X(\Omega)$ is an ordered set, then we find $\tilde{x}$ by solving equations:

$$P(X \leq \tilde{x} \geq) \geq 0.5 \text{ and } P(X \geq \tilde{x}) \geq 0.5$$

or

$$\sum_{x \leq \tilde{x}} P_X(x) \geq 0.5 \text{ and } \sum_{x \geq \tilde{x}} P_X(x) \geq 0.5$$

- Example: Two identical fair coins are tossed. Let $x =$ number of resulting heads. Then the median is $x = 1$ since $P(X \geq 1) \geq 0.5$.
- Interpretation: The data value $x = 1$ divides the data into two equal parts.

# Variance and Standard Deviation

- The **variance** of the r.v. $X$ denoted by $V(X)$ or $\sigma_X^2$, is given by:

$$V(X) = E[(X - \mu_X)^2] = \sum_{x \in X(\Omega)} (x - \mu_X)^2 P_X(x).$$

- The **standard deviation** of r.v. $X$. denoted by $\sigma_X$, is given by: $\sigma_X = \sqrt{V(X)}$.
- Standard deviation measures some type of average "variability" of the data from it mean.
- Example: Two identical fair coins are tossed. Let $X =$ number of resulting heads. Then $E(X) = 1$. We have

$$V(X) = \sum_{x \in X(\Omega)} (x - \mu_X)^2 P_X(x) = (0-1)^2 \times 1/4 + (1-1)^2 \times 1/4 + (2-1)^2 \times 1/4 = 1/2$$

and

$$\sigma_X = \frac{1}{\sqrt{2}}$$

# Moment

- The **moment** of or $r \in \mathbb{R}_{\geq 0}$ of the r.v. $X$, denoted by $(X^r)$ or $m_X^r$, is given by:

$$m_X^r = E(X^r) = \sum_{x \in X(\Omega)} x^r P_X(x).$$

  *Remark*: $\mu_X = m_X^1$.

- In general, for any function $g$ defined on $X(\Omega)$, we have

$$E[g(X)] = \sum_{x \in X(\Omega)} g(x) P_X(x).$$

- **Theorem**: $V(X) = m_X^2 - \mu_X^2$

# Binomial Distribution

- Experiment consist of $n$ trials.

  Example: Tossing a coin 15 times, 20 patients, 500 people surveyed.

- Trials are identical and each can result in one of the same two outcome ("success" and "failure"). The probability of success is $p$ and the probability of failure is $1 - p$.

  Example: Head or tail in each toss of a coin. If we are interested in observing "head", getting a head is a success and getting a tail is a failure. The probability of having a head is $p$ and the probability of getting a tail is $1 - p$.

- Trials are independent.

- Constant probability for each observation i.e. the value $p$ remains the same for each trial.

  Example: Probability of getting a head is $p$ the same each time we toss the coin.

# Binomial Probability Distribution: $X \sim \mathcal{B}(n, p)$

- $X(\Omega) = \{0, 1, 2, ..., n\}$. Parameters: $n, p$
- PMF:

$$P_X(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

- CDF:

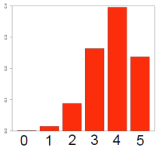
$$F_X(x) = P(X \leq x) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k}$$

- Expected value: $E(X) = np$
- Variance: $\sigma_X^2 = V(X) = np(1-p)$

# Binomial Distribution: Example

- Example: A couple, who are both carriers for a recessive disease, wish to have 5 children. They want to know the probability that they will have four healthy kids.
- We have $n = 5$. If both parents are carriers the recessive disease, the chance that the newborn has the disease is only when the newborn inherits contaminated gene from both parents. Thus, the chance that a newborn is healthy is $p = 3/4 = 0.75$.
- **Hence, the probability that the family has 4 healthy kids is**:

$$P_X(4) = P(X = 4) = \binom{5}{4} 0.75^4 \times 0.25^1 = 0.395 \text{ or } \approx 40\%$$



# Hypergeometric Distribution: $X \sim \mathcal{H}(\mathrm{H},\mathrm{K},\mathrm{n})$

- Population to be sampled consists of $N$ finite individuals, objects, or elements.
- Each individual can be characterized as a success or failure. There are $K$ successes in the population.
- A sample of size $n$ is drawn and the r.v. of interest is $X =$ number of successes.

# Binomial Distribution

- $X(\Omega) = \{\max(0, n + K - N), ..., \min(n, K)\}$. Parameters: $N, K, n$.
- PMF:

$$P_X(k) = P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

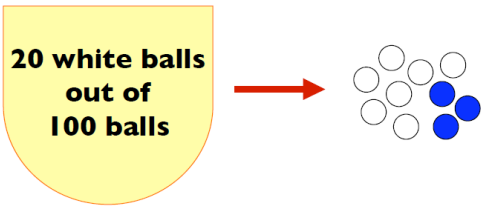
- CDF:

$$F_X(x) = P(X \leq x) = \sum_{k=0}^{x} P_X(k)$$

- Expected value: $E(X) = n\frac{K}{N}$
- Variance: $\sigma_X^2 = V(X) = n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$

# Hypergeometric Distribution: Example

- Similar in spirit to Binomial distribution, but from a finite population without replacement.




- If we randomly sample 10 balls, what is the probability that 7 or more are white?

# Poisson Process

- Suppose that events occur at random throughout an interval. Suppose further that the interval can be divided into subintervals which are so small that:
  1. The probability of more than one event occurring in the subinterval is zero.
  2. The probability of one event occurring in a subinterval is proportional to the length of the subinterval.
  3. An event occurring in any given subinterval is independent of any other subinterval.

  Then the random experiment is known as a **Poisson process**.

# Poisson Distribution: $X \sim \mathcal{P}(\lambda)$

- $X(\Omega) = \{0, 1, 2, ...\}$. Parameter: $\lambda$
- PMF:

$$P_X(x) = P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$$

  $\lambda =$ at average number of occurrences; $\alpha =$ rate of occurrence; $t =$ time unit
- CDF:

$$F_X(x) = P(X \leq x) = \sum_{k=0}^{x} P_X(x)$$

- Expected value: $E(X) = \lambda$
- Variance: $\sigma_X^2 = V(X) = \lambda$
- Safety approximate a binomial experiment when $n > 100, p < 0.01, np = \lambda < 20$.

# Poisson Distribution

- Useful in studying rare events
- If you count the number of occurrences of some event in a given time interval, e.g., number of accidents in a given day in Phnom Penh, naturally, the number has Poisson distribution.
- Poisson distribution approximates the binomial distribution when $n$ is large and $p$ is small.

# Poisson Distribution: Example

- Example: Suppose it has been observed that, on average, 180 cars per hour pass a specified point on a particular road in the morning rush hour. Due to impending roadworks it is estimated that congestion will occur closer to the city centre if more than 5 cars pass the point in any one minute. What is the probability of congestion occurring?
  - $X =$ number of cars passing the point in any minute.
  - $\lambda = 180/60 = 3$ (car per minute)
  - Find $P(X > 5)$.

# Negative Binomial or Pascal Distribution: $X \sim \mathcal{NB}(r, p)$

- Experiment consists of a sequence of trials.
- Trials are identical and each can result in one of the same two outcomes ("success" and "failure"). The probability of success is $p$ and the probability of failure is $1 - p$.
- Trials are independent
- Constant probability for each observation, i.e., the value $p$ remains the same for each trial.
- We observe this sequence until a predefined number $r$ of successes occurs.
- Then the random number of observed failures, $X$, follows the **negative binomial** or (**Pascal**) distribution.

# Negative Binomial or Pascal Distribution

- $X(\Omega) = \{0, 1, 2, ...\}$. Parameters: $r, p$
- PMF:

$$P_X(x) = P(X = x) = \binom{x + r - 1}{x}(1 - p)^x p^r$$

- CDF:

$$F_X(x) = P(X \leq x) = \sum_{k=0}^{x} P_X(k)$$

- Expected value: $E(X) = r(\frac{1-p}{p})$
- Variance: $\sigma_X^2 = V(X) = r(\frac{1-p}{p^2})$
- When $r = 1$, the Negative Binomial distribution become Geometric distribution.

# Negative Binomial Distribution: Example

- Example: Suppose a telemarketer has a 15% chance of making a sale on any given phone call. He is required to make 10 successful sales before leaving for the day. What is the probability that he needs to make more than 40 calls?
  - $X =$ number of calls that do not make a sale
  - $r = 10, p = 0.5$
  - Find $P(X > 30)$
- $P(X > 30) = 1 - P(X \leq 30) = 1 - F_X(30) = 0.9327797$

  where

$$F_X(30) = \sum_{x=0}^{30} \binom{x + 10 - 1}{x} (0.85)^x (0.15)^{10} = 0.06722026$$

# R syntax

- For Binomial distribution: `rbinom, dbinom, pbinom`
- For Hypergeometric distribution: `rhyper, dhyper, phyper`
- For Poisson distribution: `rpois, dpois, ppois`
- For Negative Binomial distribution: `rnbinom, dnbinom, pnbinom`

# Probability Density Function (PDF)

- When $X(\Omega) = \mathbb{R}$, we call the r.v. $X$ to be continuous.
- In this case, we choose $\Sigma =$ Borel $\sigma-$algebra.
- We call this function $f_X : \mathbb{R} \to \mathbb{R}$ a **probability density function (PDF)** of the random variable $X$ if for all $A \in \Sigma$,

$$P(X \in A) = \int_A f_X(x)dx.$$

- The PDF satisfies the following properties:
  - For all $x \in \mathbb{R}$, $f_X(x) \geq 0$.
  - $P(X \in \mathbb{R}) = \int_{-\infty}^{\infty} f_X(x)dx = 1$

# Probability Density Function: Example

- Example: The function

$$f_X(x) = \begin{cases} 0 \text{ if } x < 0, \\ 0.5e^{-0.5x} \text{ if } x \geq 0 \end{cases}$$

defines a probability density function for some r.v. $X$. Check that

  - $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.
  - $\int_{-\infty}^{\infty} f_X(x)dx = 1$.

# Cumulative Distribution Function (CDF)

- When $X(\Omega) = \mathbb{R}$, r.v. $X$ is said to be **continuous**. Let $f_x$ be the PDF of $X$.
- We define the **cumulative distribution function (CDF)** $F_X : \mathbb{R} \to \mathbb{R}$ by:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)dt$$

- The CDF satisfies the following properties:
  - For all $x \in \mathbb{R}, 0 \leq F_X(x) \leq 1$
  - $\lim_{x \to -\infty} F_X(x) = 0$
  - $\lim_{x \to +\infty} F_X(x) = 1$
  - $F_X(x)$ is increasing and continuous (hence continuous r.v.)
  - $f_X(x) = \frac{dF_X}{dx}(x)$

# Cumulative Distribution Function (CDF)

- Let $X$ be a continuous r.v. with PDF $f_X$. We have

$$P(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_2) = \int_{x_1}^{x_2} f_X(t)dt.$$

- For all $x \in \mathbb{R}$, $P(X = x) = 0$.
- For all $x \in \mathbb{R}$, $f_X(x) = \frac{dF_x}{dx}(x)$

# Cumulative Distribution Function (CDF)

- Example: The function

$$f_X(x) = \begin{cases} 0 \text{ if } x < 0, \\ 0.5e^{-0.5x} \text{ if } x \geq 0 \end{cases}$$

defines a PDF for a continuous r.v. $X$. Then the CDF of $X$ is

$$F_X(x) = \begin{cases} 0 \text{ if } x < 0, \\ 1 - e^{-0.5x} \text{ if } x \geq 0 \end{cases}$$

# Mean or Expected Value

- The **mean** or **expected value** of the r.v. $X$, denoted by $E(X)$ or $\mu_X$, is given by:

$$E(X) = \int_{x \in \mathbb{R}} x f_X(x) dx.$$

- Example: The function $f_X(x) = \begin{cases} 0 \text{ if } x < 0, \\ 0.5e^{-0.5x} \text{ if } x \geq 0 \end{cases}$ defines a probability density function for some r.v. $X$. Then

$$E(X) = \int_{x \in \mathbb{R}} x f_X(x) dx = \int_{-\infty}^{0} x f_X(x) dx + \int_{0}^{+infty} x f_X(x) dx.$$
$$= \int_{0}^{\infty} 0.5 x e^{-0.5x} dx = 2.$$

# Mode

- The **mode** value of the r.v. $X$, denoted by mode $(X)$, is the value $x^*$, if any, such that $f_X(x)$ is maximized ($x^*$ appears the most frequently in the data). That is,

$$x^* = \text{argmax}_{x \in X(\Omega)} f_X(x)$$

- Example: The function $f_X(x) = \begin{cases} 0 \text{ if } x < 0, \\ 0.5e^{-0.5x} \text{ if } x \leq 0.5 \end{cases}$

  defines a PDF for a continuous r.v. $X$. The mode of r.v. $X$ is mode $(X)$=1.

# Median

- The **median** value of the r.v. $X$ is the value $\tilde{x}$ that divides the data into two equal parts (hence 50%-50%). That is, if $X(\Omega)$ is an ordered set, then we find $\tilde{x}$ by solving equations:

$$P(X \leq \tilde{x}) \geq 0.5 \text{ and } P(X \geq \tilde{x}) \geq 0.5$$

or

$$\int_{-\infty}^{\tilde{x}} f_X(x)dx \geq 0.5 \text{ and } \int_{\tilde{x}}^{\infty} f_X(x)dx \geq 0.5$$

- Example: The function $f_X(x) = \begin{cases} 0 \text{ if } x < 0 \\ 0.5e^{-0.5x} \text{ if } x \geq 0 \end{cases}$ defines a PDF for a continuous r.v.$X$. Then the median of $X$ is the solution of $F_X(x) = 1 - e^{-0.5x} = 0.5$. Solving this equation, we have $\tilde{x} = 2\ln 2 \approx 1.386$.

# Variance and standard deviation

- The **variance** of the r.v. $X$, denoted by $V(X)$ or $\sigma_X^2$, is given by:

$$V(X) = E[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx.$$

- The **standard deviation** of the r.v. $X$, denoted by $\sigma_X$, is given by: $\sigma_X = \sqrt{V(X)}$. Standard deviation measures some type of average "variability" of the data from it mean.
- Example: The function $f_X(x) = \begin{cases} 0 \text{ if } x < 0, \\ 0.5e^{-0.5x} \text{ if } x \geq 0 \end{cases}$ defines a PDF for a continuous r.v. $X$. Then the median of $X$ is then $E(X) = 2$. We have

$$V(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx = \int_0^{\infty} (x - 2)^2 (0.5e^{-0.5x}) dx = 4$$

and $\sigma_X = 2$.

# Moment

- The **moment** of or $r \in \mathbb{R}_{\geq 0}$ of the r.v. $X$, denoted by $E(X')$ or $m_X^r$, is given by:

$$m_X^r = E(X^r) = \int_{-\infty}^{\infty} x^r f_X(x) dx.$$

  Remark: $\mu_X = m_X^1$.

- In general, for any function $g$ defined on $X(\Omega)$, we have

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

- Theorem: $V(x) = m_X^2 - \mu_X^2$.

# Uniform Distribution: $X \sim U[a, b]$

- The r.v $X$ is said to have a uniform distribution with parameters $a$ and $b$, written as $X \sim U[a, b]$, if its PDF is

$$f_X(x) = \begin{cases} \frac{1}{b-a} \text{ if } x \in [a, b] \\ 0 \text{ otherwise} \end{cases}$$

- CDF:

$$F_X(x) = \begin{cases} 0 \text{ if } x < 0 \\ \frac{x-a}{b-a} \text{ if } x \in [a, b] \\ 1 \text{ if } x > b \end{cases}$$

- Expected value: $E(X) = \frac{a+b}{2}$
- Variance: $\sigma_X^2 = V(X) = \frac{(b-a)^2}{12}$

# Exponential Distribution: $X \sim \mathbb{E}(\lambda)$

- the r.v. $X$ is said to have an exponential distribution with parameter $\lambda$, written as $X \sim \mathcal{E}(\lambda)$, if its PDF is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} \text{ if } x \leq 0 \\ 0 \text{ otherwise} \end{cases}$$

- CDF:

$$F_X(x) = \begin{cases} 0 \text{ if } x < 0 \\ 1 - e^{-\lambda x} \text{ if } x \geq 0 \end{cases}$$

- Expected value: $E(x) = \frac{1}{\lambda}$
- Variance: $\sigma_X^2 = V(X) = \frac{1}{\lambda^2}$

# Normal or Gaussian Distribution: $X \sim \mathcal{N}(\mu, \sigma^2)$

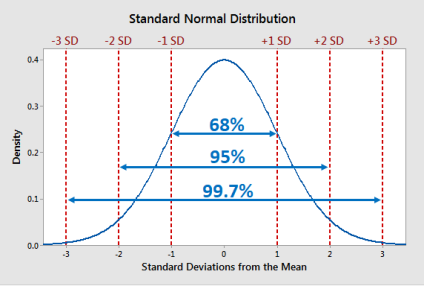
- The r.v $X$ is said to have a Gaussian distribution with parameters $\mu$ and $\sigma^2$, written as $X \sim \mathcal{N}(\mu, \sigma^2)$, if its PDF is
$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$
- CDF
$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt = \Phi(x)$$
- Expected Value: $E(X) = \mu$
- Variance: $\sigma_X^2 = V(X) = \sigma^2$
- Standard deviation: $\sigma_X = \sigma$
- Standard Normal $X \sim \mathcal{N}(0, 1)$ when $\mu = 0$ and $\sigma = 1$

# Normal or Gaussian Distribution: $X \sim \mathcal{N}(\mu, \sigma^2)$

- If $X \sin \mathcal{N}(\mu, \sigma^2)$, then

$$Z = aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$$

- $P(X \geq \mu) = P(X \leq \mu) = 0.5$
- The graph of PDF has *bell-shaped* curve
- Empirical Rule:
  - 68% of the data values lie within $\mu \pm 1\sigma$
  - 95% of the data values lie within $\mu \pm 2\sigma$
  - 99.7% of the data values lie within $\mu \pm 3\sigma$



# R syntax

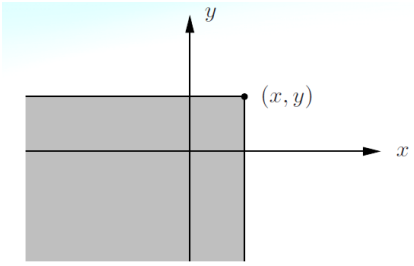
- For uniform distribution: `runif, dunif, punif`
- For exponential distribution: `rexp, dexp, pexp`
- For normal or Gaussian distribution: `rnorm, dnorm, pnorm`

# Pair of Random Variables

- A pair of r.v. $X$ and $Y$ define over the same probability space are specified by the joint CDF:

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y), \qquad x, y \in \mathbb{R}$$

- $F_{X,Y}(x,y)$ is the probability over the shaded region of $\mathbb{R}^2$
- Denote: $\mathcal{X} = X(\Omega)$ and $\mathcal{Y} = X(\Omega)$



# Pair of Random Variables

- Properties of the cdf:
  - $F_{X,Y}(x,y) \geq 0$
  - If $x_1 \leq x_2$ and $y_1 \leq y_2$ then $F_{X,Y}(x_1,y_1) \leq F_{X,Y}(x_2,y_2)$
  - $\lim_{y\to-\infty} F_{X,Y} = 0$ and $\lim_{x\to-\infty} F_{X,Y}(x,y) = 0$
  - $\lim_{y\to\infty} F_{X,Y} = F_X(x)$ and $\lim_{x\to\infty}(x,y) = F_Y(y)$
    $F_X(x)$ and $F_Y(y)$ are the **marginal cdfs** of $X$ and $Y$
  - $\lim_{x,y\to\infty} F_{X,Y}(x,y) = 1$
- $X$ and $Y$ are **independent** if for every $x$ and $y$

$$F_{X,Y}(x,y) = F_X(x)F_Y(y)$$

# Joint, Marginal, and Conditional PMFs

- Let $X$ and $Y$ be discrete random variables on the same probability space
- They are completely specified by their **joint pmf**:

$$p_{X,Y}(x,y) = P\{X = x, Y = y\}, \qquad x \in \mathcal{X}, y \in \mathcal{Y}$$

  By axioms of probability, $\sum_{x\in\mathcal{X}} \sum_{y\in\mathcal{Y}} p_{X,Y}(x,y) = 1$
- To find $p_X(x)$, the **marginal pmf** of $X$, we use the law of total probability

$$p_X(x) = \sum_{y\in\mathcal{Y}} p(x,y), \qquad c \in \mathcal{X}$$

- The **conditional pmf** of $X$ given $Y = y$ is defined as

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}, \qquad p_Y(y) \neq 0, x \in \mathcal{X}$$

# Joint, Marginal, and Conditional PMFs

- **Chain rule**: $p_{X,Y}(x,y) = p_X(x)p_{Y|X}(y|x) = p_Y(y)p_{x|y}(x|y)$
- **Independence**: $X$ and $Y$ are said to be independence if for every $(x,y) \in \mathcal{X} \times \mathcal{Y}$,

$$p_{X,Y}(x,y) = p_X(x)p_Y(y)$$

which is equivalence to $p_{X|Y}(x|y) = p_X(x)$ for every $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ such that $p_Y(y) \neq 0$

# Joint, Marginal, and Conditional PDFs

- $X$ and $Y$ are jointly **continuous** random variables if their joint cdf is continuous in both $x$ and $y$.

  In this case, we can define their **joint pdf**, provided that it exists, as the function $f_{X,Y}(x,y)$ such that

$$F_{X,Y}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)dudv, \qquad x,y \in \mathbb{R}$$

- If $F_{X,Y}(x,y)$ is differentiable in $x$ and $y$, then

$$f_{X,Y}(x,y) = \frac{\partial^2 F(x,y)}{\partial x \partial y} = \lim_{\Delta x, \Delta y \to 0} \frac{P\{x < X \leq x + \Delta x, y < Y \leq y + \Delta y\}}{\Delta x \Delta y}$$

- Properties of $f_{X,Y}(x,y)$:
  - $f_{X,Y}(x,y) \geq 0$
  - $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)dxdy = 1$

# Joint, Marginal, and Conditional PDFs

- The **marginal pdf** of $X$ can be obtained from the joint pdf via the law of total probability

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy$$

- $X$ and $Y$ are independent iff $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for every $x, y$
- **Conditional cdf and pdf**: Let $X$ and $Y$ be continuous random variables with joint pdf $f_{X,Y}(x,y)$. We wish to define $F_{Y|X}(y|X=x) = P\{Y \leq y|X=x\}$. We cannot define the above conditional probability as

$$\frac{P\{Y \leq y, X = x\}}{P\{X = x\}}$$

# Joint, Marginal, and Conditional PDFs

because both numerator and denominator are equal to zero. Instead, we define conditional probability for continuous random variables as a limit:

$$\begin{aligned} F_{Y|X}(y|x) &= \lim_{\Delta x \to 0} P\{P \leq y | x < X \leq x + \Delta x\} \\ &= \lim_{\Delta x \to 0} \frac{P\{Y \leq y, x < X \leq x + \Delta x\}}{P\{x < X \leq x + \Delta x\}} \\ &= \lim_{\Delta x \to 0} \frac{\int_{-\infty}^{y} f_{X,Y}(x,u) du \Delta x}{f_X(x) \Delta X} = \int_{-\infty}^{y} \frac{f_{X,Y}(x,u)}{f_X(x)} du \end{aligned}$$

# Joint, Marginal, and Conditional PDFs

- We then define the conditional pdf in the usual way as

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} \quad \text{if } f_X(x) \neq 0$$

- Thus

$$F_{Y|X}(y|x) = \int_{-\infty}^{y} f_{Y|X}(u|x)du$$

which shows that $f_{Y|X}(y|x)$ is a pdf for $Y$ given $X = x$, i.e.,

$$Y|\{X = x\} \sim f_{Y|X}(y|x)$$

- **Chain rule**: $f_{X,Y(x,y)} = f_X(x)f_{Y|X}(y|x)$
- **Independence**: $X$ and $Y$ are independent if $f_{X,Y} = f_X(x)f_Y(y)$ for every $(x, y)$, or equivalently, $f_{Y|X}(y|x) = f_Y(y)$.

# Covariance and Correlation Coefficients

- **Covariance** of the r.v. $X$ and $Y$ is given by:

$$COV(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X \mu_Y$$

- If $X$ and $Y$ are discrete, then:

$$COV(X, Y) = \Sigma_{x \in \mathcal{X}} \Sigma_{y \in \mathcal{Y}} (x - \mu_X)(y - \mu_Y) p_{X,Y}(x, y)$$

- If $X$ and $Y$ are continuous, then:

$$COV(X, Y) = \iint_{\mathbb{R}^2} (x - \mu_X)(y - \mu_Y) f_{X,Y}(x, y) dx dy$$

# Covariance and Correlation Coefficients

- Note that:
$$COV(X, X) = E(X^2) - \mu_X^2 = \sigma_X^2$$
- The **Correlation Coefficient** of the r.v $X$ and $Y$ is defined by
$$\rho_{X,Y} = \frac{COV(X, Y)}{\sigma_X \sigma_Y}$$

# Covariance and Correlation Coefficients

- If r.v. $X$ and $Y$ are **independent**, then

$$COV(X, Y) = \rho_{X,Y} = 0$$

- The r.v. $X$ and $Y$ is said to be uncorrelated if $\rho_{X,Y} = 0$.
- $\rho_{X,Y}$ measures the linear relationship between r.v. $X$ and $Y$.
- $-1 \leq \rho_{X,Y} \leq 1$
- $\rho_{X,Y}$ is said to be strong if $|\rho_{X,Y}|$ is close to 1 and weak if $|\rho_{X,Y}|$ is close to 0.
- $\rho_{X,Y} < 0$ refers to negative association and $\rho_{X,Y} > 0$ refers to positive association.

# Multiple of Random Variables

- An n-tuble r.v.s. $X_1, X_2, ..., X_n$ define over the same probability space are specified by the joint CDF: for $x_1, x_2, ..., x_n \in \mathbb{R}$,

$$F_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n) = P(X_1 \leq x_1, X_2 \leq x_2, ..., X_n \leq x_n)$$

- $F_{X_i}(x_i)$ denotes the marginal CDF of $X_i$:

$$F_{X_i}(x_i) = P(X_1 \leq \infty, X_2 \leq \infty, ..., X_i \leq x_i, ..., X_n \leq \infty)$$

- Independence r.v.s $X_1, X_2, ..., X_n$ are independent if for all $x_1, x_2, ..., x_n \in \mathbb{R}$, we have

$$F_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n) = \prod_{i=1}^{n} F_{X_i}(x_i)$$

# Mean and Variance of Sum Random Variables

- Suppose $X_1, X_2, ..., X_n$ are $n$ random variables.
- Let $\mu_i = E(X_i)$ and $\sigma_{ij} = COV(X_i, Y_j)$ and $\sigma_i^2 = \sigma_{ij}$
- Let $Z_n = \sigma_{i=1}^n c_i X_n$, where $c_1, c_2, ..., c_n$ are constant. Then
- $E(Z_n) = \sum_{i=1}^n c_i \mu_i$
- $V(Z_n) = \sum_{i=1}^n \sum_{j=1}^n c_i c_j \sigma_{ij} = \sum_{i=1}^n c_i^2 \sigma_i^2 + \sum_{i \leq i \neq j \leq n}^n c_i c_j \sigma_{ij}$

# Mean and Variance of Sum Random Variables

- If $X_1, X_2, ..., X_n$ are independent, then $\sigma_{ij} = 0$ and
- $E(Z_n) = \sum_{i=1}^{n} c_i \mu_i$
- $V(Z_n) = \sum_{i=1}^{n} c_1^2 \sigma_i^2$
- If $Z = aX + b$, then
  - $E(Z) = a\mu_X + b$
  - $V(Z) = a^2 \sigma_X^2$

# Distribution of Sum Random Variables

- Suppose $X_1, X_2, ..., X_n$ are $n$ independent random variables. Let $Y_n = \sum_{i=1}^{n} X_n$. Then
- If $X_i \sim \mathcal{B}(m_i, p)$, then $Y_n \sim \mathcal{B}\left(\sum_{i=1}^{n} m_i, p\right)$
- If $X_i \sim \mathcal{P}(\lambda_i)$, then $Y_n \sim \mathcal{P}\left(\sum_{i=1}^{n} \lambda_i\right)$
- If $X_i \sim \mathcal{NB}(r_i, p)$, then $Y_i \sim \mathcal{NB}\left(\sum_{i=1}^{n} r_i, p\right)$

# Distribution of Sum Random Variables

- Suppose $X_1, X_2, ..., X_n$ are $n$ independent random variables. Let $Y_n = \sum_{i=1}^{n} X_n$. then
- If $X_i \sim \mathcal{N}(\mu_n, \sigma_i^2)$, then $\sim \mathcal{N}\left(\sum_{i=1}^{n} \mu_i, \sum_{i=1}^{n} \sigma_i^2\right)$